Name _____ Period _____ Date _____

6-2 Lesson Quiz: Exponential Models

short form Savvas numbers 1, 2, 3 (items 4 and 5: full Savvas quiz only); show your work

1. In 1970, the population of a small town was 4,200. The population is decreasing at a rate of 2.4% per year.

Complete the function that models the population y in the town, t years after 1970.

$$y = _____\left(_____\right)^t$$

To the nearest thousandth of a percent, the population is decreasing about _____% per quarter.

Work (show how you found the rate per quarter):

2. The graph of an exponential model in the form $y = a \cdot b^x$ passes through the points (3, 5) and (4, 10). Which point is also on the graph?

Ⓐ (2, 0) Ⓑ (2, 1) Ⓒ (5, 15) Ⓓ (5, 20)

Work (show how you found b):

3. (Bonus) Micah invests $5,280 in an account that earns 4.2% interest, compounded monthly.

Complete the function that models the amount A in the account after t years. Write one tile in each box.

5280	5501	5298.48
$\frac{0.042}{12}$	$\frac{0.042}{6}$	0.042
$12t$	t	$\frac{t}{12}$

$$A(t) = \square\left(1+\square\right)^{\square}$$

Scoring: Questions 1 and 2, 10 points each; Question 3, bonus up to 5 points. Total out of 20. (Question numbers are the Savvas quiz numbers; items 4 and 5 are on the full Savvas quiz only.)

6-2 Lesson Quiz, short form: answer key

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1. $y = 4200(0.976)^t$**; about 0.605% per quarter.** The decay factor is $1 - 0.024 = 0.976$. For a rate per quarter, multiply the exponent by $\frac{4}{4}$: $y = 4200\left(0.976^{\frac{1}{4}}\right)^{4t} \approx 4200(0.9939452313)^{4t}$, and $100\left(1 - 0.976^{\frac{1}{4}}\right) \approx 0.6054768703\%$, so about 0.605% per quarter (round from the unrounded factor; 0.993945 would give 0.6055% and a misleading 0.606%). Dividing 2.4% by 4 (0.6%) is the common error. Savvas quiz item 1, DOK 2, Skills Review and Practice F14. Taught by Example 1 (Try It 1 is a quarterly growth rate; the packet's Practice, a car losing 12% a year, is a structural parallel). **10 points:** 2 for 4200, 3 for 0.976, 5 for 0.605% with the rewrite or $0.976^{1/4}$ shown.

2. D, (5, 20)**.** The x-values 3 and 4 are consecutive, so $b = \frac{10}{5} = 2$; then $a = \frac{5}{2^3} = 0.625$ and $y = 0.625(2)^x$, so at $x = 5$, $y = 10 \cdot 2 = 20$. Choices A and C are on the line through the two points ($y = 5x - 10$): they add 5 per step instead of multiplying by 2; B is not on the model (at $x = 2$, $y = 2.5$). Savvas quiz item 2, DOK 2, F13. Taught by Example 5 (Try It 5). **10 points:** 6 for D, 4 for the work ($b = 2$ as the ratio of the y-values).

3. Bonus: $A(t) = 5280\left(1 + \frac{0.042}{12}\right)^{12t}$**.** $P = 5280$, $r = 0.042$, $n = 12$: $A = P\left(1 + \frac{r}{n}\right)^{nt}$. Distractor tiles: $5501 \approx 5280(1.042)$ is one year at annual compounding; $5298.48 = 5280\left(1 + \frac{0.042}{12}\right)$ is the value after one month; $\frac{0.042}{6}$ and $\frac{t}{12}$ mix up the number of periods. Savvas quiz item 3, DOK 2, F67. Taught by Example 2, which the fast packet cuts; IXL YJW is the route. **Bonus 5:** 2 for 5280, 2 for $\frac{0.042}{12}$, 1 for $12t$.

Items left off this form (full Savvas quiz only; available for a retake or Form B): Savvas item 4, Steve's \$1,800 at 3.7% compounded continuously for 10 years: D, \$2,606 ($1800e^{0.37} \approx 2605.92$); DOK 2, F67; taught by Example 4, cut. Savvas item 5, Paul's \$8,000 for 6 years at 5% simple, 4.5% continuous, 4.75% yearly: 80; 89; linear; exponential; linear; exponential; DOK 2, F67; taught by Examples 2 and 4, cut. The quiz has no graphs, so nothing was left off for a figure.

IXL replacement: SmartScore 70 or above on E5S (Exponential growth and decay), 2YT (Exponential functions over unit intervals) and YJW (Compound interest: word problems) replaces this quiz with 20/20.